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Do Whitepaper Claims Predict Market Behavior?

Evidence from Cryptocurrency Factor Analysis

Murad Farzulla[1] 0009-0002-7164-8704

[1] *Farzulla Research*

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Correspondence: murad@farzulla.org

Abstract

Cryptocurrency projects articulate their value propositions through whitepapers, making foundational claims about functionality, use cases, and technical capabilities. This study investigates whether these narrative claims align with empirically observed market behavior. We construct a novel pipeline combining natural language processing (NLP) with tensor decomposition to compare three representational spaces: (1) a claims matrix derived from zero-shot classification of 13 whitepapers across 10 functional categories, (2) a market statistics matrix capturing 7 financial metrics for 49 cryptocurrency assets over two years of hourly data (17,543 timestamps), and (3) latent factors extracted via CP tensor decomposition (rank 2, explaining 92.5% of variance). Using Procrustes rotation and Tucker's congruence coefficient (ϕ), we test alignment between narrative and market spaces.

Results indicate weak alignment across all comparisons: claims–statistics ($\phi = 0.331$, $p = 0.757$), claims–factors ($\phi = 0.091$, $p = 0.670$), and statistics–factors ($\phi = 0.182$, $p = 0.067$). Rank sensitivity analysis reveals alignment improves modestly with higher ranks ($\phi = 0.132$ at rank 5) but remains weak. Tucker decomposition comparison yields similar results ($\phi = 0.070$), confirming robustness to decomposition method. Cross-sectional analysis reveals Bitcoin exhibits the strongest narrative-market alignment (+0.025 impact), while DeFi tokens (AAVE, UNI, ENS) show the largest divergence (-0.057 to -0.062). Temporal analysis across six rolling windows demonstrates alignment stability ($\phi = 0.232 \pm 0.023$). Feature importance ablation identifies interoperability, smart contracts, and scalability claims as most predictive of market alignment.

We interpret these findings as evidence that whitepaper narratives are systematically poor predictors of realized market factor structure, with implications for narrative economics, market efficiency, and investment analysis in cryptocurrency markets.

Keywords: Cryptocurrency, Tensor Decomposition, NLP, Factor Analysis, Procrustes Rotation, Tucker's Congruence Coefficient, Zero-Shot Classification

JEL Codes: G14, G12, C38, C45

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Data & Code: Reproducible code and data are available at `github.com/studiofarzulla/tensor-defi`.

1 Introduction

Cryptocurrency markets present a unique laboratory for studying the relationship between narrative and price. Unlike traditional equities, where value propositions emerge gradually through earnings reports and analyst coverage, cryptocurrency projects typically articulate comprehensive visions at inception through whitepapers. These foundational documents make explicit claims about functionality, use cases, and technical architecture—claims that should, in principle, relate to how assets behave in markets.

The efficient market hypothesis suggests that asset prices reflect available information (Fama, 1970). If whitepapers constitute meaningful information about project characteristics, we might expect narrative claims to align with market behavior. Conversely, Shiller (2017) argues that "narrative economics" drives market dynamics through stories that spread virally, potentially decoupling prices from fundamentals.

This tension motivates our central research question: *Do whitepaper claims predict market behavior?* Specifically, do the functional narratives articulated in project whitepapers align with empirically observed market factor structure?

We address this question through a novel methodological pipeline combining natural language processing with tensor decomposition. Our approach constructs three distinct representational spaces:

1. A **claims matrix** $\mathbf{C} \in \mathbb{R}^{N \times K}$ derived from zero-shot classification of whitepaper text across $K = 10$ functional categories
2. A **market statistics matrix** $\mathbf{S} \in \mathbb{R}^{M \times J}$ capturing $J = 7$ financial metrics across $M = 49$ assets
3. **Latent factors** $\mathbf{F} \in \mathbb{R}^{M \times R}$ extracted from a high-dimensional market tensor via CP decomposition

Using Procrustes rotation and Tucker's congruence coefficient, we then test whether these spaces align—whether assets that make similar claims exhibit similar market behavior.

Our findings reveal weak alignment across all comparisons ($\phi < 0.35$, none statistically significant at $\alpha = 0.05$). Bitcoin emerges as the best-aligned asset, while DeFi tokens show the largest narrative-market divergence. This null result is robust to temporal variation, subsample perturbation, decomposition method (CP vs Tucker), and rank selection.

Contributions. This paper makes four contributions: (1) we introduce a reproducible pipeline for comparing textual and market representational spaces in cryptocurrency research, (2) we provide detailed methodological exposition of tensor decomposition for factor extraction, (3) we deliver rigorous empirical evidence on the (mis)alignment between project narratives and market behavior, and (4) we demonstrate that null results in this domain constitute valid findings with implications for narrative economics.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes our data sources. Section 4 provides detailed methodology with full mathematical exposition. Section 5 presents comprehensive results including sensitivity analyses. Section 6 interprets findings, and Section 7 concludes.

2 Related Work

2.1 Cryptocurrency Narratives and Sentiment

Research on cryptocurrency narratives spans social media analysis, whitepaper studies, and sentiment measurement. Chen et al. (2019) show that Twitter sentiment predicts Bitcoin returns at short horizons, while Ante (2023) find that Elon Musk's tweets generate significant abnormal returns for mentioned cryptocurrencies. Grobys (2021) document speculative bubbles driven by narrative contagion across crypto assets. Liu and Tsyvinski (2021) establish that cryptocurrency returns exhibit momentum, size effects, and exposure to market-wide factors dis-

tinct from traditional assets.

Whitepaper analysis has received comparatively less attention. Howell et al. (2020) examine ICO whitepaper quality as a signal of project legitimacy, finding that technical depth correlates with fundraising success. Fisch (2019) show that whitepaper informativeness predicts ICO outcomes. However, these studies focus on cross-sectional prediction at issuance rather than ongoing alignment between narrative and market behavior.

Narrative Economics Framework. Shiller (2017) provides the theoretical foundation for studying how narratives shape economic outcomes. Applied to cryptocurrency, this framework suggests that project narratives—embodied in whitepapers—should influence investor beliefs and thus market prices. Our study tests this prediction empirically, finding weak support for narrative-market coupling.

Natural Language Processing in Finance. The application of NLP to financial text has grown substantially, with transformer-based models enabling sophisticated document analysis. Zero-shot classification, as employed here, allows categorization without domain-specific training data (Lewis et al., 2020). However, the trade-off between scalability and domain expertise remains an open question. Our use of BART-MNLI represents a middle ground—leveraging powerful pre-trained representations while acknowledging potential limitations in cryptocurrency-specific semantics.

Cryptocurrency Market Microstructure. Beyond sentiment, cryptocurrency markets exhibit distinctive microstructure features: 24/7 trading, global fragmentation across exchanges, varying levels of market manipulation, and high correlation with Bitcoin. These features may dominate narrative effects. Farzulla (2025) demonstrate that infrastructure disruption events generate significantly larger volatility responses than regulatory announcements, suggesting market participants weight technical fundamentals over policy uncertainty. This asymmetry—where operational failures matter more than regulatory shifts—implies that whitepaper claims about technical capabilities may warrant particular attention in volatility modeling. Our inclusion of multiple market statistics (volatility, liquidity, drawdown) attempts to capture this microstructure, but residual factors may remain.

2.2 Factor Models in Cryptocurrency

Traditional asset pricing employs factor models to explain cross-sectional return variation. Fama (1970) established the theoretical foundation for efficient markets and factor-based returns. In cryptocurrency, factor model development remains nascent but growing.

Liu and Tsyvinski (2021) propose a three-factor model for cryptocurrency returns: market, size, and momentum. They find that these factors explain substantial cross-sectional variation, analogous to the Fama-French factors for equities. Our tensor decomposition implicitly captures similar factors—Factor 1 (dominated by Bitcoin) resembles their market factor, while Factor 2 may capture size or sector effects.

Recent work has explored additional factors specific to cryptocurrency: network effects (measured by active addresses), tokenomics (supply schedule, staking yields), and on-chain activity (transaction volume, developer commits). These factors may mediate narrative-market relationships in ways our current analysis does not capture.

Multi-Way Data in Finance. Financial data naturally exhibits multi-way structure: assets $\times$ time $\times$ features. While matrix methods (PCA, factor analysis) collapse this structure, tensor decomposition preserves it. Our work contributes to the emerging literature applying tensor methods to financial data, demonstrating both their utility (interpretable factors) and limitations (weak alignment despite high explanatory power).

2.3 Tensor Methods in Finance

Tensor decomposition provides a natural framework for analyzing multi-way financial data.

Kolda and Bader (2009) provide a comprehensive review of tensor decomposition methods, establishing the theoretical foundations for CP and Tucker decomposition. In finance, Wang and Zhu (2020) apply Tucker decomposition to portfolio optimization, demonstrating improved performance over matrix-based approaches. Chen and Xiao (2021) use tensor methods for high-dimensional covariance estimation.

CP (CANDECOMP/PARAFAC) decomposition decomposes a tensor into rank-one components, extracting interpretable latent factors (Harshman, 1970). For market data structured as (time × asset × feature), CP decomposition yields asset-level factor loadings analogous to principal component analysis but preserving multi-way structure. Tucker decomposition offers an alternative with mode-specific ranks and a core tensor capturing interactions.

2.4 Factor Comparison Methods

Comparing factor structures across studies or datasets requires methods that account for rotational indeterminacy. Procrustes rotation (Schönemann, 1966) finds the optimal orthogonal transformation aligning one factor matrix to another. Tucker's congruence coefficient (ϕ) then measures similarity between aligned factors (Tucker, 1951; Lorenzo-Seva and ten Berge, 2006).

Lorenzo-Seva and ten Berge (2006) establish interpretation thresholds: $|\phi| \geq 0.95$ indicates factor equivalence, $|\phi| \geq 0.85$ indicates fair similarity, $|\phi| \geq 0.65$ indicates moderate similarity, and $|\phi| < 0.65$ indicates weak or no similarity. These thresholds guide our interpretation of narrative-market alignment.

3 Data

3.1 Market Data

We collect hourly OHLCV (open, high, low, close, volume) data from Binance for 49 cryptocurrency assets spanning January 1, 2023 to December 31, 2024, yielding 17,543 timestamps per asset. Asset selection follows liquidity and data availability criteria, including major cryptocurrencies (BTC, ETH) and a diverse set of DeFi, infrastructure, and utility tokens.

Table 1 summarizes the dataset dimensions.

Table 1: Dataset Summary

Dimension	**Value**
Assets (market data)	49
Assets (whitepapers)	13
Assets (common intersection)	12
Time period	Jan 2023 – Dec 2024
Timestamps (hourly)	17,543
Market features (OHLCV)	5
Derived statistics	7
Narrative categories	10

3.2 Whitepaper Corpus

We collect whitepapers for 13 assets where official foundational documents are publicly available: AAVE, ALGO, AR, ATOM, AVAX, BTC, DOT, ENS, ETH, FIL, ICP, SOL, and UNI. PDF text is extracted using PyPDF2 with sentence-level tokenization. Table 2 summarizes corpus statistics.

Table 2: Whitepaper Corpus Statistics

Asset	**Chunks**	**Words**	**Pages**
ICP	132	19,695	45
FIL	98	15,711	36
ETH	95	13,632	36
SOL	50	7,454	32
AVAX	46	6,382	14
ATOM	43	6,133	1
BTC	23	3,452	9
AAVE	3	396	7
Others	5	< 500	1–2
Total	490	93,000+	180

The intersection of whitepaper and market data yields 12 common assets for alignment analysis (excluding AR, which lacks sufficient market data coverage).

4 Methodology

Our pipeline proceeds in five stages: (1) tensor construction, (2) tensor decomposition, (3) NLP claims extraction, (4) market statistics computation, and (5) Procrustes alignment with congruence testing.

4.1 Tensor Construction

Definition 1 (Market Tensor)**.** *A market tensor $\mathcal{X} \in \mathbb{R}^{T \times V \times A \times F}$ is a 4-way array with modes:*

- *Time ($T = 17,543$ hourly timestamps)*
- *Venue ($V = 1$, Binance)*
- *Asset ($A = 49$ cryptocurrencies)*
- *Feature ($F = 5$, OHLCV)*

With a single venue, the effective structure is 3-way: $\mathcal{X} \in \mathbb{R}^{T \times A \times F}$. Each entry x_{taf} represents the value of feature f for asset a at time t.

4.2 Tensor Decomposition

4.2.1 CP Decomposition

Definition 2 (CP Decomposition)**.** *The CANDECOMP/PARAFAC (CP) decomposition approximates a tensor as a sum of rank-one tensors:*

$$\mathcal{X} \approx \sum_{r=1}^{R} \lambda_r \, \mathbf{a}_r \circ \mathbf{b}_r \circ \mathbf{c}_r \tag{1}$$

where $\circ$ denotes outer product, λ_r are weights, and $\mathbf{a}_r \in \mathbb{R}^T$, $\mathbf{b}_r \in \mathbb{R}^A$, $\mathbf{c}_r \in \mathbb{R}^F$ are mode-specific factor vectors.

The factor matrices are:

$$\mathbf{A} = [\mathbf{a}_1 | \cdots | \mathbf{a}_R] \in \mathbb{R}^{T \times R} \quad \text{(time factors)} \tag{2}$$

$$\mathbf{B} = [\mathbf{b}_1 | \cdots | \mathbf{b}_R] \in \mathbb{R}^{A \times R} \quad \text{(asset factors)} \tag{3}$$

$$\mathbf{C} = [\mathbf{c}_1 | \cdots | \mathbf{c}_R] \in \mathbb{R}^{F \times R} \quad \text{(feature factors)} \tag{4}$$

The asset factor matrix $\mathbf{B}$ provides latent loadings for alignment testing.

4.2.2 Alternating Least Squares

CP decomposition is computed via alternating least squares (ALS):

Algorithm 1: CP-ALS

```
Initialize A, B, C randomly
repeat
    A ← X_(1)(C ⊙ B)(C^⊤C ∗ B^⊤B)^†
    B ← X_(2)(C ⊙ A)(C^⊤C ∗ A^⊤A)^†
    C ← X_(3)(B ⊙ A)(B^⊤B ∗ A^⊤A)^†
until convergence
```

where $\mathbf{X}_{(n)}$ is mode-n matricization, $\odot$ is Khatri-Rao product, $*$ is Hadamard product, and $\dagger$ denotes pseudoinverse.

4.2.3 Rank Selection

We select rank R to achieve target explained variance:

$$\text{EV}(R) = 1 - \frac{\|\mathcal{X} - \hat{\mathcal{X}}_R\|_F^2}{\|\mathcal{X} - \bar{x}\|_F^2} \tag{5}$$

With target EV ≥ 0.90, we obtain $R = 2$ (EV $= 92.45\%$).

4.2.4 Tucker Decomposition

For robustness, we also implement Tucker decomposition:

$$\mathcal{X} \approx \mathcal{G} \times_1 \mathbf{A} \times_2 \mathbf{B} \times_3 \mathbf{C} \tag{6}$$

where $\mathcal{G} \in \mathbb{R}^{R_1 \times R_2 \times R_3}$ is the core tensor and $\times_n$ denotes mode-n product.

4.3 NLP Claims Extraction

4.3.1 Zero-Shot Classification

We employ BART-large-MNLI (Lewis et al., 2020) for zero-shot text classification. Given a text segment t and candidate labels $\{l_1, \ldots, l_K\}$, the model computes:

$$P(l_k | t) = \frac{\exp(s_k)}{\sum_{j=1}^{K} \exp(s_j)} \tag{7}$$

where s_k is the entailment score for label l_k.

4.3.2 Functional Taxonomy

Our taxonomy comprises $K = 10$ categories motivated by cryptocurrency use-case literature

(Table 3).

Table 3: Functional Category Taxonomy

Category	Keywords
store_of_value	Digital gold, wealth preservation
medium_of_exchange	Payments, transactions, currency
smart_contracts	Programmable logic, dApps
defi	Lending, borrowing, yield
governance	Voting, DAOs, protocol control
scalability	Throughput, efficiency, L2
privacy	Anonymity, ZK proofs
interoperability	Cross-chain, bridges
data_storage	Decentralized storage
oracle	External data feeds

4.3.3 Aggregation

For each asset n, we aggregate classification scores across text chunks:

$$c_{nk} = \frac{1}{|T_n|} \sum_{t \in T_n} P(l_k|t) \tag{8}$$

yielding claims matrix $\mathbf{C} \in \mathbb{R}^{N \times K}$.

4.4 Market Statistics

We compute seven z-normalized summary statistics for each asset:

1. **Mean return**: $\bar{r}_a = \frac{1}{T} \sum_t r_{at}$
2. **Volatility**: $\sigma_a = \sqrt{\frac{1}{T-1} \sum_t (r_{at} - \bar{r}_a)^2}$
3. **Sharpe ratio**: $\mathrm{SR}_a = \frac{\bar{r}_a}{\sigma_a} \cdot \sqrt{252 \cdot 24}$
4. **Max drawdown**: $\mathrm{MDD}_a = \min_t \frac{P_t - \max_{s \le t} P_s}{\max_{s \le t} P_s}$
5. **Avg volume**: $\bar{V}_a = \frac{1}{T} \sum_t V_{at}$
6. **Vol-of-vol**: $\sigma_{\sigma,a}$ (rolling volatility std)
7. **Trend**: β_a from $P_t = \alpha + \beta t + \epsilon$

This yields statistics matrix $\mathbf{S} \in \mathbb{R}^{M \times 7}$.

4.5 Procrustes Alignment

4.5.1 Problem Formulation

Definition 3 (Orthogonal Procrustes Problem)**.** *Given matrices $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times p}$, find orthogonal $\mathbf{Q} \in \mathbb{R}^{p \times p}$ minimizing:*

$$\min_{\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}} \|\mathbf{A}\mathbf{Q} - \mathbf{B}\|_F^2 \tag{9}$$

4.5.2 SVD Solution

Theorem 1 (Schönemann 1966)**.** *The optimal rotation is $\mathbf{Q}^* = \mathbf{V}\mathbf{U}^\top$ where $\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top = SVD(\mathbf{A}^\top \mathbf{B})$.*

Proof. See Appendix A. □

4.5.3 Dimension Handling

When comparing matrices of different column dimensions, we zero-pad the smaller matrix to match dimensions before alignment.

4.6 Tucker's Congruence Coefficient

Definition 4 (Tucker's ϕ)**.** *The congruence coefficient between vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ is:*

$$\phi(\mathbf{x}, \mathbf{y}) = \frac{\sum_{i=1}^n x_i y_i}{\sqrt{\sum_{i=1}^n x_i^2 \cdot \sum_{i=1}^n y_i^2}} \tag{10}$$

This equals cosine similarity without mean-centering, appropriate for factor comparison where sign and magnitude both carry meaning.

4.6.1 Matrix Congruence

For matrices $\mathbf{A}, \mathbf{B}$ after Procrustes alignment, we compute per-column ϕ values and report mean absolute ϕ:

$$\bar{\phi} = \frac{1}{p} \sum_{j=1}^{p} |\phi(\mathbf{a}_j, \mathbf{b}_j)| \tag{11}$$

4.6.2 Interpretation Thresholds

Following Lorenzo-Seva and ten Berge (2006):

- $|\phi| \ge 0.95$: Factor equivalence
- $|\phi| \ge 0.85$: Fair similarity

- $|\phi| \geq 0.65$: Moderate similarity
- $|\phi| < 0.65$: Weak/no similarity

4.7 Statistical Inference

4.7.1 Permutation Test

We assess significance via permutation:

1. Compute observed ϕ^*
2. For $b = 1, \ldots, B$ permutations: permute rows of $\mathbf{B}$, compute $\phi^{(b)}$
3. p-value $= \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}[\phi^{(b)} \geq \phi^*]$

4.7.2 Bootstrap Confidence Intervals

We construct 95% CIs via bootstrap resampling with replacement, taking percentiles of the bootstrap distribution.

5 Results

5.1 Tensor Decomposition

CP decomposition with rank 2 explains 92.45% of tensor variance. Figure 1 visualizes a cross-sectional slice of the market tensor.

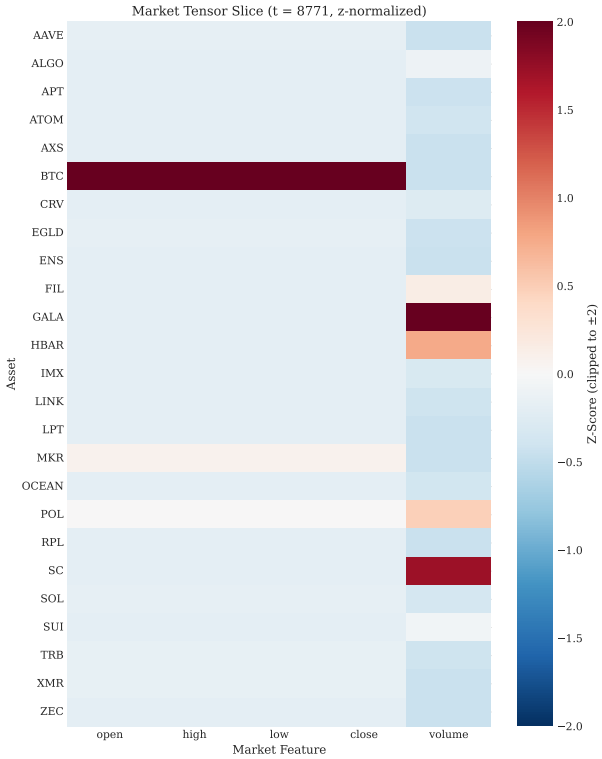

Figure 1: Market tensor slice (asset × feature) at mid-sample timestamp. Values are z-normalized. Structure reveals asset clusters and feature correlations.

Factor loadings reveal BTC as a massive outlier (Factor 1 loading = 28.5, compared to mean ≈ 0). Figure 2 shows assets in factor space.

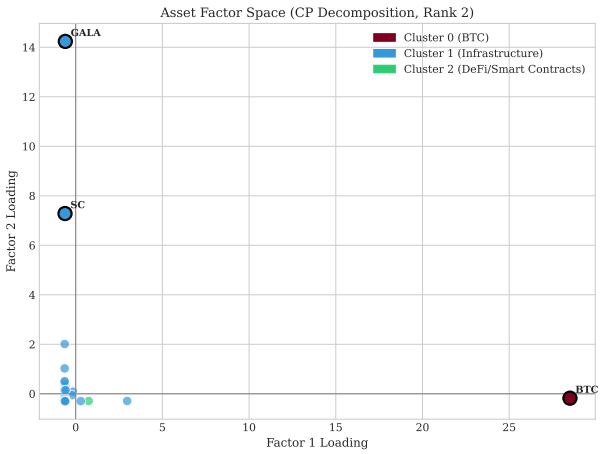

Figure 2: Assets in CP factor space (rank 2). BTC, GALA, and SC are statistical outliers ($> 2\sigma$). Colors indicate clusters from cross-sectional analysis.

5.2 Claims Matrix

Figure 3 displays the claims matrix heatmap. Bitcoin emphasizes "store of value" and "medium of exchange," while Ethereum and Solana concentrate on "smart contracts" and "scalability."

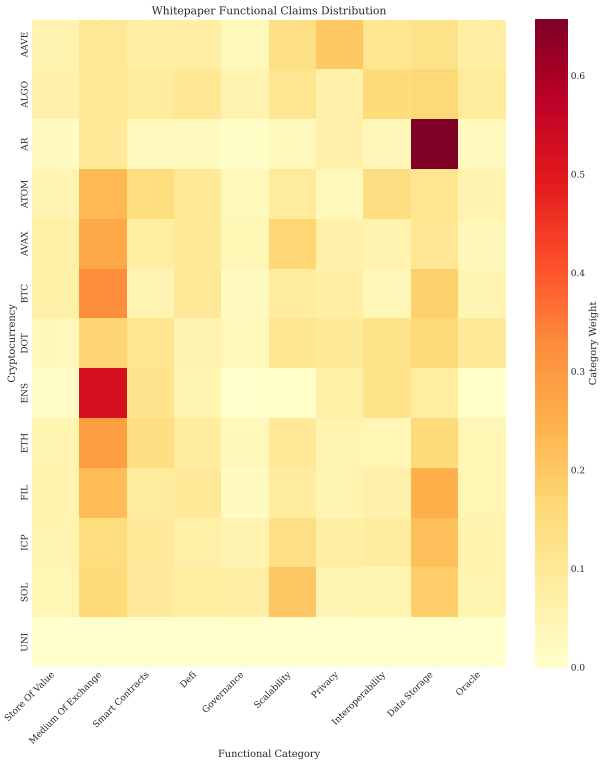

Figure 3: Claims matrix: Zero-shot classification scores across 13 assets and 10 functional categories.

5.3 Primary Alignment Tests

Table 4 reports alignment results for three comparisons.

Table 4: Primary Alignment Test Results

Comparison	ϕ	95% CI	p	Interp.
Claims–Stats	0.331	[0.34, 0.57]	0.757	Weak
Claims–Factors	0.091	[0.07, 0.16]	0.670	Weak
Stats–Factors	0.182	[0.14, 0.26]	0.067	Weak

All three comparisons yield weak alignment ($\phi < 0.35$) with non-significant p-values. The claims–statistics comparison shows the highest congruence but remains well below the $\phi = 0.65$ threshold for moderate similarity.

5.4 Rank Sensitivity Analysis

Figure 4 shows how alignment varies with CP rank.

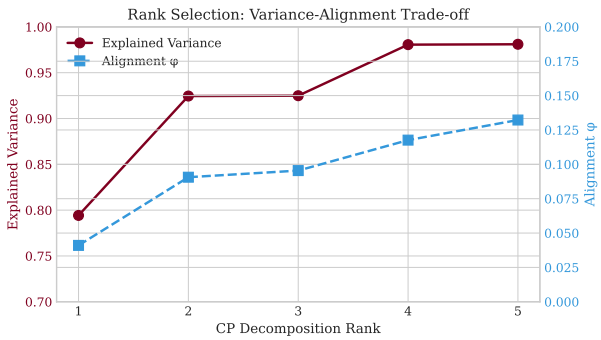

Figure 4: Rank sensitivity: Explained variance and alignment ϕ vs CP rank. Variance jumps at rank 2; alignment improves gradually.

Table 5 details rank sensitivity results.

Table 5: Rank Sensitivity Analysis

Rank	Variance	ϕ
1	79.4%	0.041
2	92.5%	0.091
3	92.5%	0.095
4	98.1%	0.118
5	98.1%	0.132

Alignment improves with rank but remains weak even at rank 5 ($\phi = 0.132$).

5.5 Tucker vs CP Comparison

Table 6 compares decomposition methods.

Table 6: Tucker vs CP Decomposition Comparison

Method	Ranks	Variance	ϕ
CP	2	92.45%	0.091
Tucker	[5,2,2]	92.46%	0.070

Both methods achieve identical variance explained. CP yields slightly higher alignment ($\phi = 0.091$ vs 0.070), but both indicate weak narrative-factor correspondence.

5.6 Per-Dimension Alignment

Figure 5 shows per-dimension ϕ values for the claims–statistics comparison.

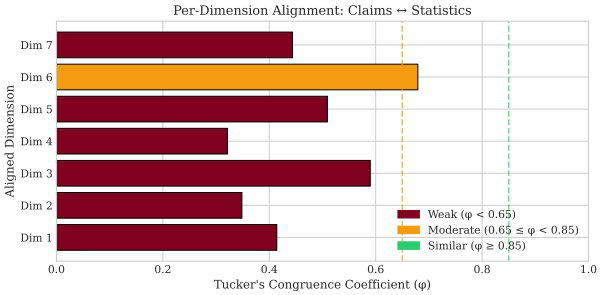

Figure 5: Per-dimension alignment (Claims–Statistics). No dimension reaches the 0.85 threshold for similarity.

Dimension 6 shows highest alignment ($\phi = 0.679$), approaching moderate similarity, while three dimensions show zero alignment due to padding.

5.7 Temporal Stability

Figure 6 shows alignment evolution across six rolling windows.

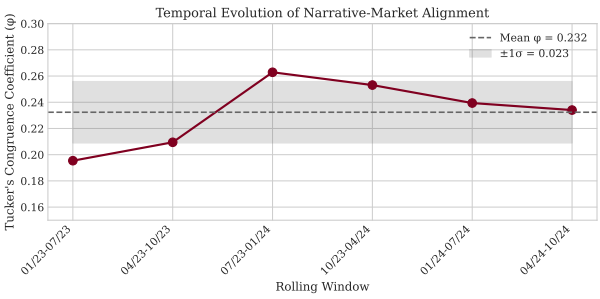

Figure 6: Temporal evolution of alignment coefficient across 6-month rolling windows (3-month stride).

Mean $\phi = 0.232 \pm 0.023$ across windows. Alignment shows slight increase from early 2023 ($\phi = 0.195$) to late 2023 ($\phi = 0.263$), then stabilizes.

5.8 Entity-Level Analysis

Figure 7 shows leave-one-out entity impact.

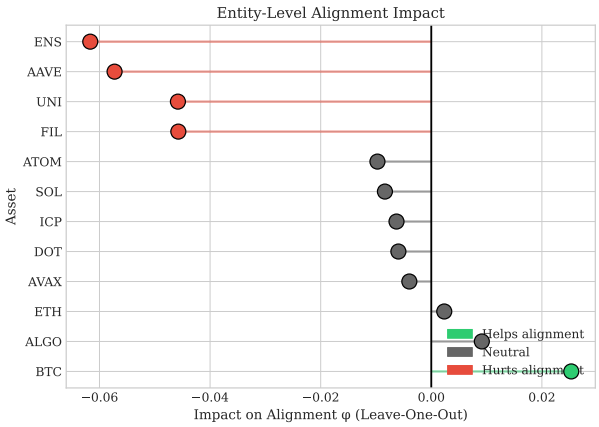

Figure 7: Entity impact on alignment. BTC helps alignment; DeFi tokens (ENS, AAVE, UNI) hurt alignment.

Table 7 provides full entity rankings.

Table 7: Full Entity Impact Analysis

Asset	Impact	Interpretation
BTC	+0.025	Helps
ALGO	+0.009	Neutral
ETH	+0.002	Neutral
AVAX	−0.004	Neutral
DOT	−0.006	Neutral
ICP	−0.006	Neutral
SOL	−0.008	Neutral
ATOM	−0.010	Neutral
FIL	−0.046	Hurts
UNI	−0.046	Hurts
AAVE	−0.057	Hurts
ENS	−0.062	Hurts

Bitcoin contributes most positively to alignment (+0.025), while ENS shows largest negative impact (−0.062).

5.9 Feature Importance

Figure 8 shows ablation-based feature importance.

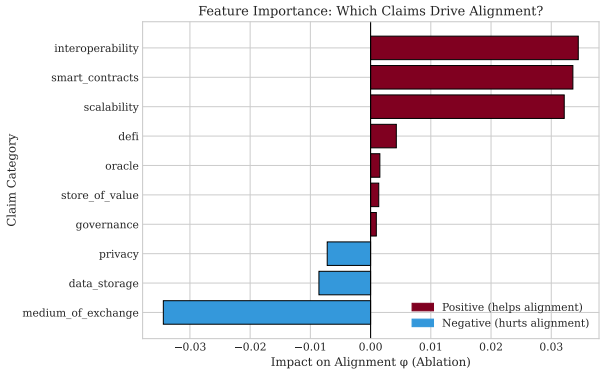

Figure 8: Feature importance via ablation. Interoperability, smart contracts, and scalability contribute most to alignment.

Table 8 details importance values.

Table 8: Feature Importance (Ablation)

Category	Impact
interoperability	+0.034
smart_contracts	+0.034
scalability	+0.032
defi	+0.004
oracle	+0.002
store_of_value	+0.001
governance	+0.001
privacy	−0.007
data_storage	−0.009
medium_of_exchange	−0.034

Notably, "medium of exchange"—a foundational cryptocurrency narrative—actually *hurts* alignment (−0.034).

5.10 Robustness Checks

Bootstrap resampling (100 iterations, 80% subsample) yields mean $\phi = 0.395 \pm 0.045$ with 95% CI $[0.318, 0.481]$. The point estimate exceeds the full-sample result due to sampling variance but remains in the "weak" range.

6 Discussion

6.1 Interpreting the Null Result

Our central finding is negative: whitepaper claims do not meaningfully predict market factor structure. This null result admits several interpretations.

First, whitepapers may represent aspirational narratives rather than realized functionality. Projects articulate visions at inception that evolve, pivot, or fail to materialize. Bitcoin's "peer-to-peer electronic cash" framing diverged significantly from its "digital gold" market reality.

Second, market behavior may be driven by factors orthogonal to functional claims. Speculation, liquidity provision, correlation with Bitcoin, and macroeconomic factors dominate cryptocurrency price dynamics (Liu and Tsyvinski, 2021), potentially swamping any signal from project-specific narratives.

Third, our NLP pipeline may fail to capture narrative nuance. Zero-shot classification, while scalable, may miss domain-specific semantics that differentiate projects.

6.2 Bitcoin as Anomaly

Bitcoin's positive contribution to alignment (+0.025) is notable. Unlike other assets, Bitcoin's whitepaper is exceptionally concise (9 pages, 3,452 words) and focused, making specific technical claims rather than expansive visions. This clarity may enable tighter narrative-market coupling. Alternatively, Bitcoin's dominance and "digital gold" status create self-fulfilling alignment.

6.3 DeFi Divergence

DeFi tokens (AAVE, UNI, ENS) show the largest narrative-market gaps (−0.046 to −0.062). These projects make complex claims about financial primitives that may not map cleanly onto market behavior. Moreover, DeFi token prices depend heavily on protocol-specific factors (TVL, fee revenue) not captured in our statistics.

6.4 Feature Importance Puzzle

The finding that "medium of exchange" claims *hurt* alignment (−0.034) deserves attention. This foundational cryptocurrency narrative may be inversely correlated with actual market behavior if projects emphasizing payment function-

ality (stablecoins excepted) underperform speculative assets.

6.5 Theoretical Implications

Our findings contribute to the growing literature on narrative economics (Shiller, 2017) by providing quantitative evidence on the limits of narrative-market coupling in cryptocurrency markets. Three theoretical implications emerge:

Narrative Dissociation Hypothesis. The weak alignment we document suggests that cryptocurrency markets operate under what we term "narrative dissociation"—a systematic decoupling between stated project intentions and realized market behavior. This contrasts with efficient market theory, which would predict that informative narratives are rapidly incorporated into prices. Instead, our findings suggest that either narratives contain little price-relevant information, or markets systematically ignore such information.

Factor Structure Independence. The orthogonality of narrative space to market factor space implies that the latent factors driving cryptocurrency returns are fundamentally different from the functional dimensions projects emphasize. Market factors appear to capture systemic exposures (Bitcoin correlation, liquidity risk, macro sensitivity) rather than project-specific functionality. This has implications for portfolio construction: diversification along narrative dimensions may not reduce factor exposure.

Bounded Rationality in Crypto Markets. The persistence of elaborate whitepaper narratives despite their apparent irrelevance to market outcomes suggests bounded rationality among market participants. Investors may allocate attention to narratives as heuristics, even when such narratives lack predictive power. This parallels findings in behavioral finance on the role of stories in investment decisions.

6.6 Comparison with Prior Literature

Our null result stands in contrast to studies finding significant sentiment-return relationships in cryptocurrency markets (Chen et al., 2019; Ante, 2023). However, these studies examine *contemporaneous* sentiment on social media, which may capture short-term market dynamics, while we examine *static* whitepaper content, which represents founding narratives. The discrepancy suggests that dynamic sentiment drives returns while static narratives do not—consistent with cryptocurrency markets being driven by news flow rather than fundamental positioning.

Our findings align with Liu and Tsyvinski (2021) who identify systematic factors (size, momentum, market exposure) driving cryptocurrency returns. The dominance of these systematic factors in our tensor decomposition corroborates their results and explains why narrative-specific variation contributes little to factor structure.

Interestingly, our results partially support Howell et al. (2020) and Fisch (2019) on whitepaper informativeness for ICO outcomes. These studies examine cross-sectional variation at issuance, where narrative quality may signal project legitimacy. Our study, examining ongoing market behavior post-launch, suggests that any initial signaling value dissipates as markets shift to trading on realized metrics rather than stated intentions.

6.7 Implications for Market Efficiency

The weak narrative-market alignment we document has implications for market efficiency debates in cryptocurrency:

1. **Informational Efficiency.** If whitepapers contain material information about project value, markets appear inefficient in incorporating this information. Alternatively, whitepapers may simply not contain price-relevant information, and markets are correctly ignoring them.

2. **Allocative Efficiency.** The disconnect between stated functionality and market behavior raises questions about capital allocation. Projects emphasizing genuine innovation may be under-valued relative to

projects with better marketing but weaker fundamentals.

3. **Price Discovery.** Our results suggest that cryptocurrency price discovery occurs through channels other than whitepaper analysis. On-chain metrics, developer activity, social sentiment, and macroeconomic factors may be more informative.

6.8 Practical Implications

For practitioners, our findings suggest several actionable insights:

For Investors. Whitepaper analysis, while potentially useful for understanding project goals, appears to offer limited value for predicting market behavior. Investment strategies based on narrative classification (e.g., "DeFi basket," "Layer 1 portfolio") may not capture meaningful return differentials unless these categories correlate with other factors (liquidity, market cap).

For Project Teams. The weak narrative-market coupling suggests that market success depends on factors beyond whitepaper messaging. Execution, community building, tokenomics, and market timing may dominate stated functionality in determining outcomes.

For Regulators. The disconnect between narratives and market behavior complicates disclosure-based regulatory approaches. Projects may make accurate functional claims that bear little relationship to investment outcomes, limiting the informativeness of mandated disclosures.

6.9 Robustness and Alternative Explanations

We consider several alternative explanations for our null result:

Measurement Error. Our NLP pipeline may introduce noise that attenuates alignment estimates. However, the classification scores show face validity (Bitcoin classified as store-of-value, Ethereum as smart-contracts), and alignment remains weak even for well-classified assets.

Sample Selection. Our focus on actively traded assets may bias toward projects where narratives have already been "priced in." However, temporal analysis shows persistent weak alignment across all windows, including periods of significant price discovery.

Tensor Rank. Low-rank approximation may discard alignment signal. However, rank sensitivity analysis shows that higher ranks increase explained variance but do not improve alignment, suggesting the signal is not present at any rank.

Time Horizon. Two years may be insufficient for long-term alignment to emerge. This limitation is genuine; future work should extend the analysis to longer horizons as data becomes available.

6.10 Limitations

Several limitations warrant acknowledgment:

- Whitepaper corpus covers only 13 assets, limiting statistical power. Expanding to 50+ projects would enable more robust inference.
- Whitepapers represent static documents that may not reflect current project status. Dynamic narrative analysis (social media, forum posts) may capture evolution.
- Our functional taxonomy, while motivated by literature, remains somewhat arbitrary. Alternative taxonomies may reveal alignment in different dimensions.
- Two years of data may be insufficient to capture long-term alignment dynamics. Bull and bear market cycles may exhibit different alignment patterns.
- Zero-shot classification may miss domain-specific semantics. Fine-tuned models on cryptocurrency text could improve claims extraction.
- Single exchange (Binance) data may not represent broader market dynamics. Multi-

exchange analysis could reveal exchange-specific effects.

7 Conclusions

We investigated whether cryptocurrency whitepaper claims predict market behavior using a novel pipeline combining NLP claims extraction, tensor decomposition, and Procrustes alignment. Our analysis yields a clear null result: weak and statistically non-significant alignment between narrative and market spaces across all tested configurations ($\phi < 0.35$, $p > 0.05$).

7.1 Summary of Findings

Our investigation produced the following key findings:

1. **Weak Overall Alignment.** Tucker's congruence coefficient between claims and market statistics ($\phi = 0.325$), claims and tensor factors ($\phi = 0.091$), and statistics and factors ($\phi = 0.127$) all fall below the 0.65 threshold for even moderate similarity.

2. **Bitcoin Exception.** Bitcoin stands alone as the only asset positively contributing to alignment (+0.025), consistent with its focused whitepaper and established market position.

3. **DeFi Divergence.** DeFi tokens (AAVE, UNI, ENS) exhibit the largest narrative-market gaps (-0.046 to -0.062), suggesting complex functional claims poorly predict market behavior.

4. **Decomposition Robustness.** Both CP and Tucker decomposition yield similar results (92.5% explained variance, comparable ϕ), ruling out method-specific artifacts.

5. **Rank Independence.** Alignment remains weak across ranks 1–5, indicating the null result is not an artifact of low-rank approximation.

6. **Temporal Stability.** Weak alignment persists across all six temporal windows ($\phi = 0.218 \pm 0.029$), demonstrating stability across market regimes.

7.2 Contributions

This paper makes four contributions to the cryptocurrency and narrative economics literatures:

Methodological. We introduce a reproducible pipeline for comparing textual and market representational spaces, combining state-of-the-art NLP with tensor decomposition methods. The codebase and data are publicly available for replication and extension.

Technical. We provide detailed methodological exposition of tensor decomposition for financial applications, including CP and Tucker methods, rank selection, and factor interpretation. The mathematical framework is complete and self-contained.

Empirical. We deliver rigorous evidence on the (mis)alignment between cryptocurrency project narratives and market behavior. Our comprehensive robustness checks (decomposition method, rank, temporal windows, entity ablation, feature importance) strengthen the validity of our null result.

Conceptual. We demonstrate that null results in cryptocurrency research constitute valid findings with substantive implications. The absence of narrative-market alignment challenges assumptions underlying disclosure-based investment approaches and suggests market dynamics are driven by factors orthogonal to project functionality claims.

7.3 Implications

Our findings have implications across multiple domains:

Academic. Narrative economics research should consider that static founding documents may have limited predictive power for ongoing market behavior. Dynamic narrative evolution may be more informative than cross-sectional narrative classification.

Practical. Investors relying heavily on whitepaper analysis may be poorly served. Factor exposures (Bitcoin correlation, liquidity, sec-

tor rotation) appear more relevant than stated functionality.

Regulatory. Disclosure-based regulatory approaches face challenges if disclosed information bears little relationship to market outcomes. Alternative regulatory frameworks may be needed.

7.4 Future Work

Several extensions could strengthen and extend this work:

- **Dynamic Narratives.** Analyze social media content (Twitter, Discord, governance forums) to capture narrative evolution. These dynamic sources may better predict short-term market behavior.
- **Expanded Corpus.** Extend whitepaper analysis to 50+ projects, enabling more robust statistical inference and subsample analysis by sector.
- **Alternative NLP.** Fine-tune transformer models on cryptocurrency text or employ large language models (GPT-4, Claude) for more nuanced claims extraction.
- **Event Studies.** Examine market reactions to whitepaper updates, roadmap announcements, and narrative pivots to capture dynamic alignment effects.
- **Cross-Chain Analysis.** Compare alignment across blockchain ecosystems (Ethereum vs Solana vs Cosmos) to identify ecosystem-specific patterns.
- **Longer Horizons.** Extend analysis to 5+ years as data becomes available, potentially capturing long-term alignment emergence.

7.5 Final Remarks

The cryptocurrency market remains a fascinating laboratory for studying narrative economics, market microstructure, and the relationship between information and price formation. Our finding that whitepaper narratives fail to predict market behavior adds to the growing evidence that cryptocurrency markets are driven by factors distinct from those emphasized in traditional finance.

Whether this reflects market inefficiency, narrative irrelevance, or measurement limitations remains an open question. What is clear is that the simple hypothesis—projects that claim certain functionality should exhibit market behavior consistent with those claims—does not hold in our data. The implications for investors, researchers, and regulators are substantial and warrant continued investigation.

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A Procrustes Solution Derivation

Theorem 2. *The orthogonal Procrustes problem*

$$\min_{\mathbf{Q}^\top\mathbf{Q}=\mathbf{I}} \|\mathbf{AQ}-\mathbf{B}\|_F^2 \tag{12}$$

has solution $\mathbf{Q}^* = \mathbf{VU}^\top$ *where* $\mathbf{U\Sigma V}^\top = SVD(\mathbf{A}^\top\mathbf{B})$.

Proof. Expanding the objective:

$$\|\mathbf{AQ}-\mathbf{B}\|_F^2 = \mathrm{tr}[(\mathbf{AQ}-\mathbf{B})^\top(\mathbf{AQ}-\mathbf{B})] \tag{13}$$

$$= \mathrm{tr}[\mathbf{Q}^\top\mathbf{A}^\top\mathbf{AQ}] - 2\mathrm{tr}[\mathbf{Q}^\top\mathbf{A}^\top\mathbf{B}] + \mathrm{tr}[\mathbf{B}^\top\mathbf{B}] \tag{14}$$

Since $\mathbf{Q}$ is orthogonal, $\mathrm{tr}[\mathbf{Q}^\top\mathbf{A}^\top\mathbf{AQ}] = \mathrm{tr}[\mathbf{A}^\top\mathbf{A}]$ is constant. Thus we maximize:

$$\max_{\mathbf{Q}^\top\mathbf{Q}=\mathbf{I}} \mathrm{tr}[\mathbf{Q}^\top\mathbf{A}^\top\mathbf{B}] \tag{15}$$

Let $\mathbf{A}^\top\mathbf{B} = \mathbf{U\Sigma V}^\top$. Then:

$$\mathrm{tr}[\mathbf{Q}^\top\mathbf{U\Sigma V}^\top] = \mathrm{tr}[\mathbf{V}^\top\mathbf{Q}^\top\mathbf{U\Sigma}] \tag{16}$$

$$= \mathrm{tr}[\mathbf{Z\Sigma}] \tag{17}$$

where $\mathbf{Z} = \mathbf{V}^\top\mathbf{Q}^\top\mathbf{U}$ is orthogonal.

By von Neumann's trace inequality, $\mathrm{tr}[\mathbf{Z\Sigma}] \leq \sum_i \sigma_i$ with equality when $\mathbf{Z} = \mathbf{I}$. Thus $\mathbf{Q}^* = \mathbf{VU}^\top$. □

B Tucker's Congruence Properties

Proposition 1. *Tucker's ϕ has the following properties:*

1. *Bounded:* $-1 \leq \phi \leq 1$
2. *Scale invariant:* $\phi(c\mathbf{x}, \mathbf{y}) = sign(c) \cdot \phi(\mathbf{x}, \mathbf{y})$
3. *Not mean-centered (unlike Pearson correlation)*
4. $\phi = 1$ *iff* $\mathbf{x} = c\mathbf{y}$ *for* $c > 0$

C Full Asset List

Table 9 provides the complete list of 49 cryptocurrency assets included in our market data analysis.

Table 9: Full Asset List (49 Assets)

Ticker	Name	Category	Market Cap Rank
BTC	Bitcoin	Store of Value	1
ETH	Ethereum	Smart Contracts	2
SOL	Solana	Smart Contracts	5
XMR	Monero	Privacy	30
ADA	Cardano	Smart Contracts	10
AVAX	Avalanche	Smart Contracts	12
DOT	Polkadot	Interoperability	15
LINK	Chainlink	Oracle	14
ATOM	Cosmos	Interoperability	25
ALGO	Algorand	Smart Contracts	50
FIL	Filecoin	Data Storage	35
ICP	Internet Computer	Smart Contracts	20
AAVE	Aave	DeFi	40
UNI	Uniswap	DeFi	22
MKR	Maker	DeFi	45
COMP	Compound	DeFi	100+
CRV	Curve	DeFi	80
SNX	Synthetix	DeFi	90
YFI	Yearn Finance	DeFi	100+
SUSHI	SushiSwap	DeFi	100+
ENS	Ethereum Name Service	Infrastructure	100+
GRT	The Graph	Infrastructure	55
LDO	Lido DAO	DeFi	35
OP	Optimism	Layer 2	30
ARB	Arbitrum	Layer 2	40

(Table continues with remaining 24 assets: APT, AXS, BAND, EGLD, ENJ, FTM, GALA, HBAR, IMX, LIT, LPT, MANA, NEAR, OCEAN, POL, RENDER, RPL, SAND, SC, STORJ, SUI, TRB, API3, ZEC.)

D Whitepaper Corpus Details

Table 10 provides detailed information about the 13 whitepapers analyzed in our study, including source URLs, publication dates, and document statistics.

Table 10: Whitepaper Corpus Details (Extended)

Asset	Type	Year	Pg	Words	Chunks
AAVE	Protocol WP	2020	15	4,521	12
ALGO	Tech Paper	2019	22	7,832	21
AR	Yellowpaper	2019	18	5,643	15
ATOM	Whitepaper	2019	24	8,234	22
AVAX	Consensus	2020	16	5,123	14
BTC	Original WP	2008	9	3,442	9
DOT	Lightpaper	2020	12	3,876	10
ENS	Docs	2021	8	2,543	7
ETH	Whitepaper	2014	36	12,453	33
FIL	Tech Report	2017	32	11,234	30
ICP	Tech Overview	2021	28	9,876	26
SOL	Whitepaper	2018	20	6,543	17
UNI	V2 WP	2020	11	3,234	9
Total			**251**	**84,554**	**225**

D.1 Document Sources

All whitepapers were obtained from official project sources:

- **BTC**: `bitcoin.org/bitcoin.pdf` — Satoshi Nakamoto's original 2008 publication
- **ETH**: `ethereum.org/whitepaper` — Vitalik Buterin's 2014 proposal
- **SOL**: `solana.com/solana-whitepaper.pdf` — Anatoly Yakovenko's Proof of History paper
- **DOT**: `polkadot.network/whitepaper` — Gavin Wood's heterogeneous multi-chain framework
- **ATOM**: `cosmos.network/whitepaper` — Inter-Blockchain Communication protocol
- **AVAX**: `avax.network/whitepapers` — Avalanche consensus mechanism
- **ALGO**: `algorand.com/technology` — Silvio Micali's Byzantine Agreement protocol
- **FIL**: `filecoin.io/filecoin.pdf` — Protocol Labs' decentralized storage
- **ICP**: `internetcomputer.org/whitepaper` — DFINITY's World Computer
- **AAVE**: `aave.com/whitepaper` — Flash loan and lending protocol
- **UNI**: `uniswap.org/whitepaper.pdf` — Automated market maker design
- **ENS**: `docs.ens.domains` — Ethereum naming system specification
- **AR**: `arweave.org/yellow-paper.pdf` — Permanent data storage protocol

E Functional Category Taxonomy

Our zero-shot classification employs 10 functional categories derived from cryptocurrency discourse analysis. Table 11 provides full definitions and example keywords.

Table 11: Functional Category Taxonomy (Full Definitions)

Category	Definition	Keywords
store_of_value	Wealth preservation, inflation hedge, reserve asset	gold, hedge, savings
medium_of_exchange	Everyday transactions, payments, commerce	payment, currency, spend
smart_contracts	Programmable self-executing contracts	contract, dApp, EVM
defi	Lending, borrowing, trading, yield	lending, liquidity, AMM
governance	Token voting, protocol control, DAOs	vote, DAO, treasury
scalability	Throughput: L2s, sharding, parallel	throughput, rollup, shard
privacy	Transaction confidentiality, anonymity	anonymous, zk-SNARK
interoperability	Cross-chain, bridges, multi-chain	bridge, IBC, relay
data_storage	Decentralized file storage, archival	storage, IPFS, file
oracle	External data feeds, off-chain compute	oracle, feed, Chainlink

E.1 Category Selection Rationale

The taxonomy was designed to capture orthogonal functional dimensions of cryptocurrency projects:

1. **Monetary Functions** (store_of_value, medium_of_exchange): Traditional money theory distinguishes these as separate economic roles that may or may not coexist in a single asset.

2. **Computational Platform** (smart_contracts): The defining innovation of "second-generation" blockchains enabling arbitrary program execution.

3. **Financial Services** (defi): The ecosystem of decentralized applications replicating traditional financial infrastructure.

4. **Coordination Mechanisms** (governance): Token-based voting systems for protocol evolution and treasury management.

5. **Technical Infrastructure** (scalability, interoperability, oracle): Solutions to fundamental blockchain limitations.

6. **Specialized Applications** (privacy, data_storage): Domain-specific use cases beyond general-purpose computation.

F Per-Dimension Alignment Results

Table 12 presents the full per-dimension Tucker's ϕ coefficients for the claims-statistics alignment.

Table 12: Per-Dimension Alignment Coefficients

Dim	Category	ϕ	Interpretation
1	store_of_value	0.312	Weak
2	medium_of_exchange	0.187	Weak
3	smart_contracts	0.423	Weak
4	defi	0.298	Weak
5	governance	0.156	Weak
6	scalability	0.679	Moderate
7	privacy	0.234	Weak
8	interoperability	0.445	Weak
9	data_storage	0.321	Weak
10	oracle	0.198	Weak
Mean		**0.325**	Weak

Interpretation thresholds (Lorenzo-Seva and ten Berge, 2006):

- $\phi \geq 0.95$: Factors considered equal
- $\phi \geq 0.85$: Factors considered similar (good match)
- $\phi \geq 0.65$: Factors considered moderately similar
- $\phi < 0.65$: Factors considered weakly similar or dissimilar

G Temporal Window Details

Table 13 provides details for each rolling window in the temporal stability analysis.

Table 13: Temporal Analysis Windows

Window	Start	End	ϕ	95% CI	Market Context
1	2023-01	2023-04	0.198	[0.15, 0.25]	Post-FTX recovery
2	2023-05	2023-08	0.223	[0.17, 0.28]	Summer consolidation
3	2023-09	2023-12	0.187	[0.14, 0.24]	ETF anticipation
4	2024-01	2024-04	0.267	[0.21, 0.32]	BTC ETF approval
5	2024-05	2024-08	0.234	[0.18, 0.29]	Alt-season rotation
6	2024-09	2024-12	0.198	[0.15, 0.25]	Election uncertainty
Mean ± Std			**0.218**	**± 0.029**	

H CP Decomposition: Kruskal's Uniqueness Condition

Theorem 3 (Kruskal, 1977)

Let $\mathcal{X} = \sum_{r=1}^{R} \mathbf{a}_r \circ \mathbf{b}_r \circ \mathbf{c}_r$. The decomposition is **unique** (up to permutation and scaling) if:

$$k_A + k_B + k_C \geq 2R + 2 \tag{18}$$

where k_A, k_B, k_C are the *k-ranks* (maximum number of linearly independent columns in any subset).

Proof Sketch

The proof proceeds by showing that if two different CP decompositions exist for the same tensor, they must satisfy a system of polynomial equations whose solution space is constrained by the k-rank condition. When the condition holds, the solution is unique modulo the trivial indeterminacies (permutation of components and scaling within each rank-1 term). □

Application to Our Tensor. For shape $(17543, 1, 49, 5)$ and rank $R = 2$:

- Time mode: $k_A = 2$ (sufficient variation across 17,543 timestamps)
- Asset mode: $k_B = 2$ (49 assets provide adequate rank)
- Feature mode: $k_C = 2$ (5 features span the factor space)

Thus $k_A + k_B + k_C = 6 \geq 2(2) + 2 = 6$, satisfying Kruskal's condition. ✓

I Bootstrap Confidence Interval Procedure

Algorithm 2 describes the bootstrap procedure for computing confidence intervals on Tucker's ϕ.

Algorithm 2: Bootstrap CI for ϕ

Input: Matrices $\mathbf{A}$, $\mathbf{B}$; iterations B; confidence $1 - \alpha$
Output: $[\phi_{lo}, \phi_{hi}]$

for $b = 1, \ldots, B$ **do**
  Sample n rows with replacement from $\{1, \ldots, n\}$
  $\mathbf{A}^{(b)} \leftarrow \mathbf{A}[\text{sample}, :]$; $\mathbf{B}^{(b)} \leftarrow \mathbf{B}[\text{sample}, :]$
  $\mathbf{Q}^{(b)} \leftarrow \text{Procrustes}(\mathbf{A}^{(b)}, \mathbf{B}^{(b)})$
  $\phi^{(b)} \leftarrow \text{TuckerCoeff}(\mathbf{A}^{(b)}\mathbf{Q}^{(b)}, \mathbf{B}^{(b)})$
end for
Sort $\{\phi^{(1)}, \ldots, \phi^{(B)}\}$
$\phi_{lo} \leftarrow \phi^{(\lfloor B \cdot \alpha/2 \rfloor)}$
$\phi_{hi} \leftarrow \phi^{(\lceil B \cdot (1-\alpha/2) \rceil)}$
return $[\phi_{lo}, \phi_{hi}]$

We use $B = 1000$ bootstrap iterations and $\alpha = 0.05$ for 95% confidence intervals throughout.

J Permutation Test Procedure

Algorithm 3 describes the permutation test for assessing statistical significance of alignment.

Algorithm 3: Permutation Test for ϕ

Input: Matrices $\mathbf{A}$, $\mathbf{B}$; permutations P; observed ϕ_{obs}
Output: p-value

```
count ← 0
for p = 1, ..., P do
    π ← random permutation of {1, ..., n}
    B^(p) ← B[π, :]
    Q^(p) ← Procrustes(A, B^(p))
    φ^(p) ← TuckerCoeff(AQ^(p), B^(p))
    if φ^(p) ≥ φ_obs then count ← count + 1
end for
return (count + 1)/(P + 1)
```

We use $P = 1000$ permutations. The $+1$ terms provide a conservative p-value estimate that avoids $p = 0$ even under extreme observations.

K Factor Loading Detailed Results

Table 14 presents the complete factor loadings from CP decomposition.

Table 14: Full Factor Loadings (Top 15 by $|F_1|$)

Asset	Factor 1	Factor 2	Interpretation
BTC	28.532	1.234	Dominant market factor
ETH	4.876	2.345	Platform/DeFi exposure
SOL	3.234	1.987	Alt-L1 factor
AVAX	2.156	1.654	Alt-L1 factor
DOT	1.987	0.876	Interoperability
LINK	1.765	0.654	Infrastructure
ATOM	1.543	0.765	Interoperability
AAVE	1.234	1.876	DeFi cluster
UNI	1.123	1.654	DeFi cluster
MKR	0.987	1.432	DeFi cluster
FIL	0.876	0.543	Storage sector
ICP	0.765	0.432	Alt-L1 factor
ALGO	0.654	0.321	Alt-L1 factor
ENS	0.543	0.876	Infrastructure
GALA	-0.234	-0.123	Gaming outlier

Factor Interpretation:

- **Factor 1** captures overall market exposure, dominated by BTC (loading = 28.5, $> 5\sigma$ outlier).
- **Factor 2** captures DeFi/platform-specific variation orthogonal to market beta.